Mohomad Fazil

fazilfrd9@gmail.com
0729755982
linkedin.com/in/mohomad-fazil
github.com/Fazilfareed
No. 8B, Rathwaththa Mawatha, Badulla

PROFESSIONAL PROFILE

Computer Science undergraduate at University of Jaffna (CGPA: 3.78/4.0) with a strong foundation in algorithms, machine learning, networking, and software development. Currently researching optimization-enhanced hybrid ML/DL models for intrusion detection in IoT systems. Experienced in full-stack development using the MERN stack and mentoring students as a Python Practical Assistant, with demonstrated problem-solving ability through competitive programming and academic projects.

EDUCATION

Bachelor of Science (Hons) in Computer Science Jun 2022 – Present
University of Jaffna — CGPA: 3.78/4.0 (73 Credits)

G.C.E. Advanced Level (Mathematics Stream) 2020
Badulla Central College — Results: 1A, 2C — Z-score: 1.0644

PROJECTS

1. A Particle Swarm Optimization Enhanced Hybrid Machine Learning and Deep Learning Technique for Intrusion Detection in Internet of Things (Academic Research) 2025 – Present

- Designed and implemented a hybrid Machine Learning and Deep Learning framework for Intrusion Detection in IoT environments.
- Applied Particle Swarm Optimization (PSO) for feature selection to enhance detection accuracy and reduce model complexity.
- Evaluated model performance using the RT_IOT2022 dataset with dataset-driven experimentation and comparative analysis.
- Analyzed intrusion detection metrics including accuracy, precision, recall, F1-score, and false positive rate.
- Focused on improving detection efficiency and robustness for real-world IoT security scenarios.

2. Bandwidth Optimization Through Data Caching (Networking Project) 2026 – Present

- Designed a caching-based traffic reduction strategy to minimize redundant data transmission.
- Conducted performance analysis focusing on throughput improvement and latency reduction.

- Evaluated network efficiency under varying load conditions.

3. PaySync – Installment-Based Finance Management System (MERN Stack) 2025

- Developed a full-stack web and mobile application for managing installment-based purchases.
- Implemented role-based access control for hierarchical staff and client management.
- Designed loan approval workflows and installment tracking mechanisms.
- Built RESTful APIs and integrated MongoDB for scalable data management.

4. Inventory Management System (PHP & MySQL) 2024

- Developed a department-level inventory tracking system with structured database design.
- Implemented advanced search functionality and dynamic data filtering.
- Integrated PDF and Excel export features for reporting and documentation.

4. To-Do Mobile Application (React Native) 2024

- Built a cross-platform mobile application for task creation and management.
- Implemented state management and persistent storage for user tasks.
- Designed a clean and responsive user interface.

5. IoT Defected Cookie Detection System (Edge Computing Project) 2024

- Designed an AI-based architecture for identifying defective cookies in a manufacturing process.
- Configured Docker containers and Ubuntu (WSL) environment for model deployment.
- Incorporated edge computing principles for real-time defect detection.

6. Bluetooth Controlled RC Car (Arduino Project – Design Phase) 2023

- Designed a Bluetooth-controlled RC car system architecture using Arduino.
- Simulated circuit logic and control flow using TinkerCAD.
- Planned hardware-software integration for remote navigation control.

7. Tic Tac Toe Desktop Application (Java) 2023

- Developed a GUI-based desktop game using Java in Eclipse IDE.
- Implemented game logic, input validation, and win-condition algorithms.
- Applied object-oriented programming principles for structured design.

WORK & EXPERIENCE

Python Practical Assistant – University of Jaffna Feb 2025 – Apr 2025, Feb 16, 2026 – Present

- Conduct practical sessions for students
- Assist students in debugging and programming fundamentals

Web Development Support – shop1.lk

- Assisted in development and maintenance of WordPress website
- Provided support for e-commerce management

SKILLS & COMPETENCIES

Programming Languages: Python, Java, JavaScript, PHP
Databases: MongoDB, MySQL
Machine Learning & AI: Machine Learning, Deep Learning, Intrusion Detection Systems (IoT), Edge Computing
Networking & Security: Networking, Cybersecurity, System Architecture Design
Tools & Platforms: VS Code, Eclipse, Git, GitHub, WordPress, TinkerCAD, Docker, Ubuntu (WSL)
Frameworks & Libraries: MERN Stack (MongoDB, Express.js, React, Node.js), React Native, Arduino libraries
Soft Skills: Problem Solving, Critical Thinking, Leadership, Teamwork, Academic Mentoring, Project Management, Time Management, Communication, Analytical Thinking, Attention to Detail

VOLUNTEERING & LEADERSHIP

Membership Development Committee Member, IEEE Student Branch University of Jaffna Feb 2023 – Feb 2024

- Contributed to member engagement initiatives and branch development activities.
- Assisted in organizing IEEE-related events and outreach programs.

Volunteer, Career Fair 2025 2025

- Assisted in event coordination, and participant support.

Team Lead – Decoration Team, DCS Got Talent 2024

- Led the decoration team and coordinated planning and execution of event setup.
- Managed task delegation and ensured timely completion of responsibilities.

AWARDS & ACHIEVEMENTS

- First Runner-Up – UOJ Coders v4.0 2025
- Participant – IEEE Xtreme 19.0 2025
- Participant – IEEE Xtreme 17.0 2023
- Participant – YarlITHub Coding Competition 2024
- Participant – Yarl Insight 2.0 (Summer School on MERN Stack) 2025
- Participant – Sri Lanka Student Workshop in Computer Science 2023, 2025
- Inter-Faculty Carrom Tournament – Team Championship Winner 2023

CERTIFICATIONS & E-LEARNING

Certificate Courses – Cisco Networking Academy

- IT Essentials

Certificate Courses – University of Moratuwa

- Fullstack Web Development

Certificate Courses – Microsoft

- GitHub for Beginners

Certificate Courses – SLIIT

- AI/ML for Beginners

Certificate Courses – Hacking.lk

- Ethical Hacking

LANGUAGES

English (Fluent) Sinhala (Native) Tamil (Fluent)